

Subsystem Specification

Hull & structure

Provide a stable, unsinkable, modular catamaran platform that a 4-year-old can help assemble, carrying the electronics box, thruster pods, mast and DUPLO deck.

Document	BOATY-SSS-HUL
Issue	Issue A (for review)
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Parents	SRS Issue D, ADD Issue C, ICD Issue B
Content	19 allocated SRS requirements → 25 subsystem requirements (19 M, 5 S, 1 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Provide a stable, unsinkable, modular catamaran platform that a 4-year-old can help assemble, carrying the electronics box, thruster pods, mast and DUPLO deck.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Six hull segments with closed-cell foam cores • Two crossbeams with M3 rail • Electronics box, saddle and glands • Camera hood • DUPLO deck plate • Mast tube and socket • Carry handle • All structural fasteners	• Thruster pods (PRP) • Anything inside the box (PWR, HLM, MCP) • Flag, hoop, beacon, labels (REC)

1.3 Breakdown

Item	Description
HUL-1 Hull segments x6	Bow, mid and stern per hull; ≤ 200 mm long; foam-filled PETG shells
HUL-2 Crossbeams x2	Box-section beams with a 10 mm M3 hole grid on top
HUL-3 Electronics box	Clip-lock box, saddle, 4 x PG7 glands, key dock
HUL-4 Deck & mast	DUPLO plate, mast tube, socket, handle
HUL-5 Camera hood	Printed spray hood and window on the box front

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue C	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue B	Interface definitions (all listed in section 3 apply in full)

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-16 Hull segment joint	internal	Owner	\$8
IF-17 Pod mount	Propulsion	Conforms	\$8
IF-18 Box & glands	Power	Owner	\$8
IF-19 Mast socket	Helm	Owner	\$8
IF-20 DUPLO deck	Operator & crew	Owner	\$8

4. Parent requirements

Primary (19): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
CAM-006	S	The lens shall be shielded from spray and cleanable on the bank without tools.	HUL-D19
REC-001	M	The boat shall stay afloat, carrying its maximum payload, with the electronics box flooded and any two hull segments...	HUL-D05, HUL-D06
MEC-001	M	The boat shall be a twin-hull catamaran no more than 650 mm long, 400 mm wide and 500 mm high (including mast).	HUL-D01
MEC-003	M	The boat shall carry at least 300 g of payload on the deck with at least 50 mm of freeboard.	HUL-D07
MEC-004	M	Static heel shall be 10° or less with 300 g at the deck edge. The boat shall not capsize in full-speed turns or in 100...	HUL-D04, HUL-D08, HUL-D09
MEC-005	M	Every printed part shall fit a 200 × 200 × 200 mm build volume (Ultimaker).	HUL-D02
MEC-006	M	Wetted and exposed printed parts shall be PETG or ASA. Hull segments shall be filled with closed-cell foam.	HUL-D05, HUL-D10
MEC-007	M	Hull segments, thruster pods, mast, electronics box and DUPLO deck shall be removable and refittable by hand, without...	HUL-D02, HUL-D14, HUL-D16, HUL-D25
MEC-009	M	The deck plate shall carry LEGO DUPLO-compatible studs. Genuine bricks shall grip firmly and be removable by a...	HUL-D17
MEC-012	M	The electronics box shall let in no water after 30 min at 0.3 m depth. Every cable entry shall go through a gland.	HUL-D15
MEC-014	S	The boat shall fit in a car boot and have a carry handle for one adult hand.	HUL-D21
MEC-015	S	Fasteners shall be stainless steel or plastic.	HUL-D12
ENV-003	M	The boat shall operate in waves up to 100 mm.	HUL-D09
ENV-004	S	The boat should keep operating in light rain. Heavy rain is outside the operating envelope.	HUL-D13
ENV-006	C	Exposed parts shall keep their function over two seasons of outdoor use.	HUL-D11
CHD-001	M	No child-handled part shall fit wholly inside the EN 71-1 small-parts cylinder.	HUL-D22
CHD-002	M	Child-handled parts shall have rounded edges (radius at least 0.5 mm) and no pinch points.	HUL-D23
CHD-003	M	The battery, electronics and propellers shall not be reachable without tools or an adult action.	HUL-D24
CHD-004	M	'Child tier' assembly (hull segments, pods, deck, flag) shall need no tools and be doable with small hands.	HUL-D14, HUL-D25

Supporting (5): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

CAM-005, MEC-008, MEC-013, PWR-010, ENV-002

5. Subsystem requirements

Requirements are numbered HUL-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Geometry and mass

ID	Requirement	Pri	Ver	Stage	Trace
HUL-D01	Overall length shall be ≤ 620 mm and beam ≤ 360 mm. Height including the mast shall be ≤ 500 mm. Margins of 30 mm / 40 mm under MEC-001 for build tolerance.	M	I	BENCH	MEC-001
HUL-D02	Each hull shall comprise three segments (bow, mid, stern), each ≤ 200 mm long and printable in a $200 \times 200 \times 200$ mm volume.	M	I	BENCH	MEC-005, MEC-007
HUL-D03	Hull and structure mass (segments, beams, deck, box, mast, fasteners) shall be ≤ 1000 g.	M	T	BENCH	MEC-002
HUL-D04	Hull centreline spacing shall be ≥ 270 mm. Drives heel stability; see section 6.	M	A	BENCH	MEC-004

5.2 Buoyancy and stability

ID	Requirement	Pri	Ver	Stage	Trace
HUL-D05	Closed-cell foam shall fill $\geq 90\%$ of each segment's internal volume.	M	I	BENCH	MEC-006, REC-001
HUL-D06	With the box flooded and any two segment shells breached, remaining buoyancy shall be $\geq 1.5 \times$ design mass (2.1 kg).	M	T	POOL	REC-001
HUL-D07	At design mass with 300 g on the deck, draft shall be ≤ 35 mm and freeboard ≥ 50 mm.	M	T	POOL	MEC-003
HUL-D08	Static heel with 300 g at the deck edge shall be $\leq 10^\circ$.	M	T	POOL	MEC-004
HUL-D09	The boat shall not capsize in full-speed (1.5 m/s) on-the-spot turns or in 100 mm waves with 300 g at the deck edge.	M	D	POOL	MEC-004, ENV-003

5.3 Materials and durability

ID	Requirement	Pri	Ver	Stage	Trace
HUL-D10	Exposed and wetted printed parts shall be PETG or ASA, with ≥ 3 perimeters, ≥ 4 top/bottom layers and a hull shell ≥ 1.6 mm thick.	M	I	BENCH	MEC-006
HUL-D11	Exposed parts shall be UV-stable (ASA, or PETG with UV coating) and keep their function after two seasons.	C	I	LAKE	ENV-006
HUL-D12	Fasteners shall be A2/A4 stainless steel or plastic. No plain steel.	S	I	BENCH	MEC-015
HUL-D13	Deck and box lid shall shed rain: no water inside the box after a 10 min drip test, and no pooling on the deck.	S	T	BENCH	ENV-004

5.4 Modularity and interfaces

ID	Requirement	Pri	Ver	Stage	Trace
HUL-D14	Segment joints shall conform to IF-16 (captive thumb-screws, dowels).	M	T	BENCH	IF-16, MEC-007, CHD-004
HUL-D15	The electronics box shall conform to IF-18 and pass 30 min at 0.3 m without ingress. Every cable entry uses a gland.	M	T	BENCH	IF-18, MEC-012
HUL-D16	The box saddle shall place the box lid ≥ 150 mm above the waterline and allow adult removal without tools.	M	I	BENCH	MEC-007, COM-002
HUL-D17	The DUPLO deck shall conform to IF-20.	M	T	BENCH	IF-20, MEC-009

ID	Requirement	Pri	Ver	Stage	Trace
HUL-D18	The mast socket shall conform to IF-19, with the GNSS ≥ 150 mm from power wiring.	M	I	BENCH	IF-19, MEC-013
HUL-D19	The camera hood shall overhang the lens by ≥ 15 mm and leave the window reachable with a cloth, no tools.	S	I	POOL	CAM-006
HUL-D20	The camera mount shall hold the lens ≥ 120 mm above the waterline, level to $\pm 3^\circ$, facing forward.	S	I	BENCH	CAM-005

5.5 Handling and child safety

ID	Requirement	Pri	Ver	Stage	Trace
HUL-D21	A carry handle within ± 50 mm of the centre of gravity shall carry $3 \times$ design mass. With the mast removed, the boat shall fit a $650 \times 400 \times 250$ mm envelope.	S	T	BENCH	MEC-014
HUL-D22	No child-handled part (segments, deck plate, thumb-screws, pod shells) shall fit wholly inside the EN 71-1 small-parts cylinder. Fasteners the crew handles are captive.	M	T	BENCH	CHD-001
HUL-D23	Child-handled edges shall have ≥ 0.6 mm chamfer or radius. Joints close face-to-face with no finger-trap geometry.	M	I	BENCH	CHD-002
HUL-D24	The box and battery bay shall open only by an adult action: latches needing ≥ 40 N or a two-handed release.	M	T	BENCH	CHD-003
HUL-D25	The crew's assembly steps (segments, pods, deck, flag) shall need no tools and ≤ 20 N finger force.	M	D	BENCH	CHD-004, MEC-007

6. Design analysis: buoyancy and stability

Hand estimates that size HUL-D04 to D09. They are checked against CAD volumes and then pool tests.

Quantity	Working	Result
Hull volume	$2 \times 600 \times 90 \times 100 \text{ mm} \times C_b 0.6$	6.5 L
Design mass	Boat 1.8 kg + payload 0.3 kg	2.1 kg
Draft	Displaced 2.1 L / waterplane 756 cm ² (Cwp 0.7)	$\approx 28 \text{ mm} (\leq 35 \text{ mm})$
Reserve buoyancy	6.5 L vs 2.1 L	3.1×
Two segments lost (worst case, no foam credit)	$6.5 \times 4/6$	$4.3 \text{ L} = 2.1 \times \text{design mass} (\geq 1.5 \times)$
Transverse BM (catamaran)	$2 \cdot A_{wp} \cdot (s/2)^2 / V$, $s = 270 \text{ mm}$	$\approx 0.66 \text{ m}$
Righting moment	$\Delta \cdot g \cdot GM \cdot \sin 1^\circ$, $GM \approx BM$	$\approx 0.24 \text{ N} \cdot \text{m per degree}$
Heel, 300 g at 180 mm	0.53 N·m / RM per degree	$\approx 2.2^\circ (\leq 10^\circ)$

A catamaran this wide is very stiff. The practical capsizing risk is a wave or wake over a low bow, which is why HUL-D09 is demonstrated rather than calculated.

7. Design constraints and budgets

- ADD DD-14: printed foam-filled segments; PVC pontoons are the fallback
- Build volume 200 mm cube (Ultimaker)
- PETG default material; ASA for high-UV parts if needed

Budget	Allocation
Mass allocation	$\leq 1000 \text{ g}$ (estimate 930 g)
Cost allocation	£5 (box, glands); filament from stock
Power	None

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	-	-	-	-	-	0
BENCH	8	1	9	1	-	19
POOL	3	1	1	-	-	5
LAKE	-	-	1	-	-	1

Ref	Open item
TBC-11	Final hull lines from CAD (IF-16)
TBC-13	Box selection and saddle (IF-18)
TBC-14	DUPLO stud size by test print (IF-20)

Subsystem Specification Propulsion

Turn helm commands into bidirectional thrust on two clip-on pods, quietly, safely and without being defeated by weed.

Document	BOATY-SSS-PRP
Issue	Issue A (for review)
Date	28 September 2026
Status	For review by the project owner
Parents	SRS Issue D, ADD Issue C, ICD Issue B
Content	7 allocated SRS requirements → 15 subsystem requirements (9 M, 6 S, 0 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Turn helm commands into bidirectional thrust on two clip-on pods, quietly, safely and without being defeated by weed.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Two thruster pods (motor, prop, shroud/guard, strut, dovetail, latch) • Two bidirectional ESCs (in the box) • Pod connectors and motor leads	• Motor power (PWR) • Motor commands and speed control (HLM) • Transom mount features (HUL)

1.3 Breakdown

Item	Description
PRP-1 Thruster pod ×2	2204-2208 outrunner running submerged; printed prop in a full shroud
PRP-2 ESC ×2	20-30 A, BLHeli_S/AM32, bidirectional, DShot
PRP-3 Pod connector ×2	Keyed 3-pole IP68

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue C	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue B	Interface definitions (all listed in section 3 apply in full)

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-05 Motor commands	Helm	Conforms	§6
IF-06 Motor power	Power	Conforms	§6
IF-17 Pod mount	Hull & structure	Owner	§8

4. Parent requirements

Primary (7): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
NAV-001	M	The boat shall steer by differential thrust from two thrusters, and be able to go forward and in reverse.	PRP-D01, PRP-D02, PRP-D06, PRP-D15
FS-008	M	Helm failure: if motor commands stop arriving from the helm, the motors shall stop within 1 s.	PRP-D07, PRP-D08
MEC-008	S	Swapping a thruster pod shall take under 2 min. Pod connectors shall be waterproof and keyed.	PRP-D12
MEC-010	M	Propellers shall be guarded so that an 8 mm diameter probe cannot touch a blade.	PRP-D09
MEC-011	S	Thruster pods shall include weed-shedding features: a swept leading edge and a guard shape that sheds weed.	PRP-D10
ENV-002	M	The boat shall make at least 0.5 m/s headway into a 5.4 m/s wind (top of Beaufort 3).	PRP-D02
ENV-005	S	The boat shall keep operating in sparse surface weed, recovering through FS-005 and FS-006.	PRP-D10, PRP-D11

Supporting (8): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

NAV-002, NAV-003, FS-006, PWR-009, CHD-001, CHD-002, CHD-003, CHD-004

5. Subsystem requirements

Requirements are numbered PRP-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Performance

ID	Requirement	Pri	Ver	Stage	Trace
PRP-D01	Two identical pods, port and starboard, shall each give forward and astern thrust on command.	M	D	POOL	NAV-001
PRP-D02	Combined static forward thrust shall be ≥ 4 N, and astern ≥ 1.5 N. Sized from the section 6 estimate: ≈ 1.1 N needed for 0.5 m/s into 5.4 m/s wind, with a margin of 3.5.	M	T	BENCH	ENV-002, NAV-001
PRP-D03	The boat shall reach 1.0 m/s in calm water at $\leq 50\%$ throttle.	M	T	POOL	NAV-002
PRP-D04	Combined electrical input at 1.0 m/s cruise shall be ≤ 10 W.	S	T	POOL	PWR-009
PRP-D05	Continuous current shall be ≤ 8 A per ESC at full throttle, measured.	M	T	BENCH	IF-06

5.2 Control interface and failure behaviour

ID	Requirement	Pri	Ver	Stage	Trace
PRP-D06	The ESCs shall accept commands per IF-05 (DShot300, 3D mode; PWM fallback).	M	T	BENCH	IF-05, NAV-001
PRP-D07	Each ESC shall stop its motor ≤ 1 s after command pulses stop.	M	T	BENCH	FS-008, V-09
PRP-D08	After signal loss, an ESC shall not restart until it sees a neutral command.	M	T	BENCH	FS-008

5.3 Guarding and weed

ID	Requirement	Pri	Ver	Stage	Trace
PRP-D09	Each prop shall be fully shrouded, so that an 8 mm probe cannot touch a blade from any direction.	M	T	BENCH	MEC-010
PRP-D10	Guard and strut leading edges shall be swept back $\geq 30^\circ$, with no forward-facing hooks. Inlet bars shall run at $\geq 45^\circ$ to the flow. Prop blades shall have swept leading edges.	S	I	BENCH	MEC-011, ENV-005
PRP-D11	In sparse surface weed, a pod fouled at the inlet shall be clearable by ≤ 3 reverse bursts (FS-006) in ≥ 2 of 3 trials.	S	T	LAKE	ENV-005

5.4 Mechanical and handling

ID	Requirement	Pri	Ver	Stage	Trace
PRP-D12	Pods shall mount per IF-17 and be swappable in < 2 min, tool-free, with keyed waterproof connectors.	S	T	BENCH	MEC-008, IF-17
PRP-D13	Each pod shall weigh ≤ 75 g including the motor.	S	T	BENCH	MEC-002
PRP-D14	Pod shells and latches are child-handled parts (HUL-D22/D23 apply). The prop shall not be reachable while the pod is fitted or unfitted.	M	I	BENCH	CHD-001, CHD-002, CHD-003
PRP-D15	Motors are consumables: rinse and dry after each session. Expected life ≥ 10 h submerged before replacement.	S	A	LAKE	NAV-001

6. Design analysis: thrust and power

Case	Working	Result
Water drag at 0.5 m/s	$\frac{1}{2}\rho v^2 \cdot S \cdot C_t$, $S = 0.18 \text{ m}^2$, $C_t = 0.012$, $\times 1.5$ form	0.41 N
Wind drag, 5.4 m/s head wind	$\frac{1}{2}\rho_{\text{air}}(5.9 \text{ m/s})^2 \cdot 0.035 \text{ m}^2 \cdot 1.1$	0.82 N
Needed for ENV-002	sum	$\approx 1.2 \text{ N}$
Specified thrust (PRP-D02)	$\geq 4 \text{ N}$ combined	margin $\approx 3.3\times$
Drag at 1.0 m/s cruise	with wave-making $\times 1.3$	$\approx 1.4 \text{ N} \rightarrow 1.4 \text{ W}$ at the water
Electrical at cruise	overall efficiency $\approx 15\%$	$\approx 9 \text{ W}$ (PRP-D04 $\leq 10 \text{ W}$)
Small printed props in shrouds are inefficient. The 15% figure is deliberately pessimistic and gets replaced by bench thrust-stand data.		

7. Design constraints and budgets

- ADD T3/T5: submerged brushless, differential thrust, no rudder
- Airboat pod is a future swap-in module (same IF-17)

Budget	Allocation
Mass allocation	$\leq 200 \text{ g}$ (estimate 180 g)
Cost allocation	£22 (motors £12, ESCs £10)
Power	$\approx 8 \text{ W}$ at cruise (estimate); $\leq 16 \text{ A}$ total continuous

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	-	-	-	-	-	0
BENCH	8	-	2	-	-	10
POOL	2	1	-	-	-	3
LAKE	1	-	-	1	-	2

Ref	Open item
TBC-05	ESC DShot and 3D-mode support (IF-05)
TBC-06	Measured motor current (IF-06)
TBC-12	Pod connector part number (IF-17)
V-09	ESC stop on signal loss $\leq 1 \text{ s}$

Subsystem Specification

Power

Store and distribute energy safely. Give motor power only when an adult has inserted the magnetic key. Measure what's used.

Document	BOATY-SSS-PWR
Issue	Issue A (for review)
Date	28 September 2026
Status	For review by the project owner
Parents	SRS Issue D, ADD Issue C, ICD Issue B
Content	10 allocated SRS requirements → 19 subsystem requirements (18 M, 1 S, 0 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)

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1. Scope

1.1 Purpose

Store and distribute energy safely. Give motor power only when an adult has inserted the magnetic key. Measure what's used.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• 3S Li-ion 18650 pack with BMS • Main fuse and main switch • Magnetic arming key, reed switch and MOSFET motor-rail switch • Power module (V/I sense) • 5 V buck for the mission computer • Harness and connectors	• Battery charger (owned; off-boat) • Helm battery monitoring configuration (HLM)

1.3 Breakdown

Item	Description
PWR-1 Battery pack	3 × 18650 ≥ 3000 mAh, 3S BMS, XT60
PWR-2 Protection & switching	20 A fuse, IP67 main switch, key switch
PWR-3 Conversion & sensing	Power module, 5 V 3 A buck, rail divider
PWR-4 Harness	Silicone wire, XT60/XT30, keyed plugs

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue C	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue B	Interface definitions (all listed in section 3 apply in full)

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-06 Motor power	Propulsion	Owner	\$6
IF-07 Helm power & sensing	Helm	Owner	\$6
IF-08 Mission computer power	Mission computer	Owner	\$6
IF-18 Box & glands	Hull & structure	Conforms	\$8

4. Parent requirements

Primary (10): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
MOD-003	M	Arming shall require two deliberate adult actions: the arming switch on the boat set to ARM, and arming confirmed at...	PWR-D07, PWR-D08
PWR-001	M	Propulsion and avionics shall run from one removable 3S (nominal 11.1 V) lithium battery of 25 to 50 Wh.	PWR-D01
PWR-002	M	The battery shall be protected against over-discharge, over-current and short circuit (BMS or equivalent).	PWR-D02
PWR-003	M	A main fuse shall sit within 50 mm of the battery positive, rated no more than twice the maximum continuous current.	PWR-D05
PWR-004	M	A waterproof main power switch shall be operable without opening the electronics box.	PWR-D06
PWR-005	M	Avionics shall have a separate regulated 5 V supply. A full reverse-to-forward thrust step shall not reset or brown...	PWR-D10, PWR-D11, PWR-D12
PWR-006	M	Battery voltage and current shall be measured, logged and used for state of charge and failsafes.	PWR-D13
PWR-007	M	The battery shall be removed from the boat for charging, and charged only off-boat.	PWR-D03, PWR-D04
PWR-008	M	All power connectors shall be polarised and keyed, with no two same-type connectors doing different jobs.	PWR-D14, PWR-D15
PWR-011	S	A power budget shall be kept up to date alongside the design.	PWR-D17

Supporting (7): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

MOD-002, PRE-003, PRE-007, MEC-012, PWR-009, ENV-004, CHD-003

5. Subsystem requirements

Requirements are numbered PWR-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Energy storage

ID	Requirement	Pri	Ver	Stage	Trace
PWR-D01	The battery shall be 3S1P Li-ion 18650. Cells ≥ 3000 mAh and ≥ 10 A continuous; 30-36 Wh nominal.	M	I	BENCH	PWR-001
PWR-D02	A 3S BMS shall provide over-discharge (2.5-2.8 V/cell), over-current (trip ≤ 25 A), short-circuit and balance protection.	M	T	BENCH	PWR-002
PWR-D03	The pack shall be in a rigid sleeve with XT60 and a balance lead, removable from the box by an adult in < 30 s without tools.	M	D	BENCH	PWR-007
PWR-D04	The boat shall have no charging port. Charging is by external balance charger only.	M	I	BENCH	PWR-007

5.2 Protection and switching

ID	Requirement	Pri	Ver	Stage	Trace
PWR-D05	A 20 A blade fuse shall sit ≤ 50 mm from the battery positive.	M	I	BENCH	PWR-003, IF-06
PWR-D06	The main switch shall be IP67, outside the box, and either rated ≥ 20 A or control a solid-state switch rated ≥ 30 A.	M	I	BENCH	PWR-004
PWR-D07	The motor rail shall be switched by the magnetic key per IF-06: reed switch + MOSFET, off when the key is absent.	M	T	BENCH	MOD-003, IF-06
PWR-D08	Motor-rail voltage shall be divided to the helm's second voltage input so arming is refused below 9.0 V.	M	T	BENCH	MOD-003, PRE-007, V-13
PWR-D09	At power-up the motor rail shall be off unless the key is present. The helm is disarmed regardless (HLM-D04).	M	T	BENCH	MOD-002

5.3 Conversion and sensing

ID	Requirement	Pri	Ver	Stage	Trace
PWR-D10	The helm shall be powered from the main bus before the key switch, through its own regulator (IF-07).	M	I	BENCH	PWR-005, IF-07
PWR-D11	The mission computer shall have a dedicated 5.1 V ≥ 3 A buck (IF-08).	M	T	BENCH	PWR-005, IF-08
PWR-D12	Helm and mission-computer supplies shall stay in regulation through a full reverse-to-forward thrust step at 12.6 V and at 9.0 V.	M	T	BENCH	PWR-005
PWR-D13	Battery voltage and current shall be measured per IF-07 ($\pm 1\%$ V, $\pm 5\%$ I after calibration).	M	T	BENCH	PWR-006, PRE-003

5.4 Harness

ID	Requirement	Pri	Ver	Stage	Trace
PWR-D14	Connectors: XT60 battery, XT30 per ESC, JST-XH balance, a keyed 2-pin for the mission computer. No connector type used for two functions. All polarised.	M	I	BENCH	PWR-008
PWR-D15	Wire: 16 AWG battery to bus, 18 AWG to ESCs, silicone insulated. Joints soldered and heat-shrunk. Strain relief at every gland.	M	I	BENCH	PWR-008, MEC-012
PWR-D16	No exposed live conductor outside the box.	M	I	BENCH	CHD-003

5.5 Budgets and thermal

ID	Requirement	Pri	Ver	Stage	Trace
PWR-D17	A power budget (docs/budgets/power.csv) shall be kept current, with $\geq 30\%$ margin at the design endurance.	S	A	BENCH	PWR-011
PWR-D18	Usable energy ≥ 25 Wh, supporting ≥ 40 min at the budgeted cruise load.	M	A	BENCH	PWR-009
PWR-D19	The battery shall sit ≥ 30 mm from the ESCs. Box internal temperature ≤ 50 °C after 60 min at 30 °C ambient in sun.	M	T	BENCH	PWR-010

6. Design analysis: power budget

Load	W
Helm + GNSS	0.8
Mission computer (avg)	1.8
Beacon	0.3
ESC idle	0.2
Motors at cruise	8.0
Total at cruise	11.1
Usable energy (3S, 3.0 Ah, 80%)	25.9 Wh
Endurance estimate	≈ 140 min (≥ 40 min required, ≥ 60 target)
Margin at 40 min	250%

Circuit: battery → fuse → main switch → power module → bus. The bus feeds the helm (own regulator), the 5 V buck (MCP) and, through the key-switched MOSFET, the ESCs. See ADD figure 6.

7. Design constraints and budgets

- ADD DD-09: 3S Li-ion 18650 + BMS
- ADD DD-10: magnetic key removes motor power independently of software
- Charging only off-boat (PWR-007)

Budget	Allocation
Mass allocation	≤ 260 g (estimate 250 g)
Cost allocation	£31 (cells + BMS £15, fuse/box share £2, buck £3, key switch £4, IP67 main switch £4, sensing/wiring £3)
Energy	≈ 32 Wh nominal, ≈ 26 Wh usable

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	-	-	-	-	-	0
BENCH	8	1	8	2	-	19
POOL	-	-	-	-	-	0
LAKE	-	-	-	-	-	0

Ref	Open item
TBC-06	Motor current measurement → fuse and MOSFET rating
TBC-07 / V-13	Arming gated on motor-rail voltage
PWR-OI-1	Closed: IP67 main switch added to the ADD BOM (Issue C)

Subsystem Specification Helm

Navigate, enforce the fence and run every native failsafe on an independent RTOS processor, whatever else has failed.

Document	BOATY-SSS-HLM
Issue	Issue A (for review)
Date	28 September 2026
Status	For review by the project owner
Parents	SRS Issue D, ADD Issue C, ICD Issue B
Content	37 allocated SRS requirements → 38 subsystem requirements (33 M, 3 S, 2 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Navigate, enforce the fence and run every native failsafe on an independent RTOS processor, whatever else has failed.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Flight controller hardware (F405-class) • ArduPilot Rover firmware (third party) and the controlled parameter set • GNSS + compass module and mast cabling • Fence and mission storage • Dataflash / SD logging	• Everything that sends commands (MCN via IF-02, MCP via IF-04) • Motor drives (PRP) • Power sources (PWR)

1.3 Breakdown

Item	Description
HLM-1 Flight controller	F405-class, ChibiOS; IMU, baro, UARTs, outputs, ADCs
HLM-2 Firmware	ArduPilot Rover, pinned version
HLM-3 Parameter set	params/boaty-mk1.parm (controlled)
HLM-4 GNSS/compass	u-blox M10 + magnetometer on the mast

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue C	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue B	Interface definitions (all listed in section 3 apply in full)

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-02 Command & telemetry	Mission Control	Conforms	\$4
IF-04 Companion link	Mission computer	Conforms	\$4
IF-05 Motor commands	Propulsion	Owner	\$6
IF-07 Helm power & sensing	Power	Conforms	\$6
IF-09 Beacon drive	Recovery & signalling	Owner	\$6
IF-19 Mast socket	Hull & structure	Conforms	\$8
IF-22 GNSS & compass	internal	Owner	\$6

4. Parent requirements

Primary (37): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
MOD-001	M	The system shall implement exactly these boat modes: DISARMED, HOLD, MANUAL, AUTO and RTL, as defined in section 2.3.	HLM-D05
MOD-002	M	On power-up the boat shall enter DISARMED, with both motors inhibited.	HLM-D04
MOD-004	M	The system shall refuse to arm unless every pre-arm check (PRE-001 to PRE-008) passes, and shall state the failing...	HLM-D08
MOD-005	M	When an AUTO mission completes, the boat shall enter RTL. On reaching home (within the waypoint acceptance radius) it...	HLM-D06
MOD-008	C	The boat shall disarm automatically after 60 s in HOLD at home with no operator input, and announce it.	HLM-D07
PRE-001	M	GNSS shall have a 3D fix with at least 8 satellites and HDOP no greater than 1.5.	HLM-D09
PRE-002	M	A geofence shall be loaded and enabled, and home shall lie inside the inclusion fence and outside all exclusion zones.	HLM-D10
PRE-003	M	Battery state of charge shall be at least 80%.	HLM-D11
PRE-004	M	The heading sensor shall report healthy and calibrated.	HLM-D12
PRE-007	M	The independent safety layer (SAF-001) shall report healthy.	HLM-D13
PRE-008	M	Logging shall be active on the helm and Mission Control, each with at least 1 GB free.	HLM-D02, HLM-D14
NAV-002	M	Cruise speed in calm water shall be 1.0 m/s $\pm$ 0.2 m/s.	HLM-D15
NAV-003	M	Speed through the water shall be limited to 1.5 m/s in every mode.	HLM-D16
NAV-004	M	In AUTO the boat shall pass within the acceptance radius of each waypoint. The radius is configurable from 1 to 10 m,...	HLM-D17
NAV-005	S	Cross-track error between waypoints shall be no more than 3 m RMS in wind up to 3 m/s.	HLM-D18
NAV-006	S	In HOLD with station-keeping, the boat shall stay within 5 m of the hold point in wind up to 3 m/s.	HLM-D18
NAV-007	M	In RTL the boat shall navigate to home without entering any exclusion zone or leaving the inclusion fence.	HLM-D19
NAV-008	S	Above 0.3 m/s the heading estimate shall use GNSS course as well as the magnetometer, so a disturbed compass cannot...	HLM-D20
FEN-001	M	The system shall support one inclusion polygon of 3 to 70 vertices and up to 10 exclusion zones, each a polygon or...	HLM-D10, HLM-D21
FEN-004	M	The fence shall be enforced in every armed mode, including MANUAL.	HLM-D22
FEN-005	M	On a fence breach the boat shall enter RTL within 1 s of the breach.	HLM-D23
FEN-006	M	If the boat stays outside the inclusion fence for more than 30 s, or gets more than 10 m outside it, the motors shall...	HLM-D24
FEN-007	M	Fence and exclusion zones shall not be changed while the boat is armed.	HLM-D25
FS-001	M	Low battery: when estimated state of charge falls to 35%, the boat shall enter RTL. At 15% it shall continue RTL at...	HLM-D26
FS-002	M	Link loss in MANUAL: with no command link for 2 s the boat shall enter HOLD, and after 10 s it shall enter RTL.	HLM-D27
FS-003	M	Link loss in AUTO: the boat shall continue the mission. If the link is still lost after 60 s it shall enter RTL.	HLM-D27
FS-004	M	Position loss: if the GNSS fix is lost, HDOP exceeds 2.5 or the position estimate is unhealthy for 3 s, the motors...	HLM-D28

SRS	Pri	Requirement (abridged)	Met by
FS-005	M	Stuck detection: if commanded thrust is at least 50% and ground speed stays below 0.1 m/s for 5 s, the boat shall...	HLM-D29
FS-007	M	Mission computer failure: if the helm receives no mission computer heartbeat for 3 s, it shall complete the loaded...	HLM-D30
FS-011	M	When several failsafes are active, the most conservative response shall win: motors stopped over RTL over continuing...	HLM-D31
FS-012	M	Every failsafe event shall be logged, and announced at Mission Control within 2 s if the link is up.	HLM-D32
SAF-001	M	Fence and failsafe functions shall run on a processor separate from the mission computer and Mission Control. That...	HLM-D01, HLM-D02, HLM-D03
SAF-002	M	Those functions shall keep working with the mission computer, Mission Control and the radio link all failed.	HLM-D30, HLM-D33
SAF-003	M	The mission computer and Mission Control shall command the helm only through the helm interface (SWE-002). They shall...	HLM-D34
COM-007	C	An independent manual-override radio link (e.g. ExpressLRS) may be added.	HLM-D37
MEC-013	M	The GNSS and compass shall be mounted at least 150 mm from motor and power wiring.	HLM-D35
LOG-001	M	The helm shall log position, attitude, mode, commands, battery and failsafe events at 5 Hz or faster.	HLM-D36

Supporting (22): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

MOD-003, MOD-006, PRE-005, NAV-001, FS-008, FS-009, FS-010, SAF-007, MIS-001, MIS-002, MIS-003, MIS-004, MIS-005, MIS-006, DET-004, MC-014, REC-004, PWR-005, PWR-006, PWR-010, SWE-008, ENV-002

5. Subsystem requirements

Requirements are numbered HLM-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Hardware and firmware

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D01	The helm shall run a pinned ArduPilot Rover stable release (≥ 4.5) built for the chosen board. Version and hash are recorded in the repository.	M	I	BENCH	SAF-001, SWE-006
HLM-D02	The flight controller shall provide: ≥ 2 free UARTs (GNSS, companion), ≥ 3 DShot-capable outputs, ≥ 2 analogue battery inputs, I2C, IMU and baro, and log storage for ≥ 2 h at the configured rate (microSD preferred; see HLM-OI-1).	M	I	BENCH	SAF-001, PRE-008, V-01
HLM-D03	No project-written code (including Lua) shall run on the flight controller. It runs ChibiOS only.	M	I	BENCH	SAF-001, DD-13
HLM-D04	On power-up the helm shall be disarmed with motor outputs at neutral. Auto-arming shall be disabled.	M	T	BENCH	MOD-002

5.2 Modes

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D05	Only STEERING, HOLD, LOITER, AUTO, RTL and GUIDED (weed-shedding only) shall be used. No RC mode switch is configured.	M	I	SIM	MOD-001, DD-12
HLM-D06	At mission end the helm shall RTL and then hold at home. The mission-done behaviour is set to hold as a backstop.	M	S	SIM	MOD-005, V-04
HLM-D07	Auto-disarm after 60 s at home is performed by Mission Control (MCN-D12). The helm shall accept it.	C	S	SIM	MOD-008

5.3 Pre-arm checks

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D08	All ArduPilot arming checks shall be enabled, and failures reported as STATUSTEXT (MCN translates them).	M	S	SIM	MOD-004
HLM-D09	Arming shall need a 3D fix with HDOP ≤ 1.5 . The ≥ 8 satellite check is done by Mission Control (MCN-D15).	M	T	BENCH	PRE-001
HLM-D10	Arming shall need the fence enabled and loaded, with home inside it.	M	S	SIM	PRE-002, FEN-001
HLM-D11	Arming shall need $\geq 80\%$ battery capacity remaining.	M	S	SIM	PRE-003
HLM-D12	Arming shall need calibrated, consistent compass and EKF.	M	T	BENCH	PRE-004
HLM-D13	Arming shall be refused with motor-rail voltage < 9.0 V (key out).	M	T	BENCH	PRE-007, MOD-003, V-13
HLM-D14	Arming shall need logging available.	M	T	BENCH	PRE-008

5.4 Navigation

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D15	Cruise speed shall be set to 1.0 m/s and tuned to ± 0.2 m/s.	M	T	POOL	NAV-002
HLM-D16	Waypoint and RTL speeds shall be ≤ 1.5 m/s. Maximum throttle shall be limited so full throttle in STEERING gives ≤ 1.5 m/s in calm water.	M	T	POOL	NAV-003
HLM-D17	Waypoint acceptance radius shall default to 3 m (configurable 1-10 m).	M	S	SIM	NAV-004

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D18	Steering and speed controllers shall be tuned to ≤ 3 m RMS cross-track error and ≤ 5 m loiter error in 3 m/s wind.	S	T	LAKE	NAV-005, NAV-006
HLM-D19	RTL shall plan around exclusion zones inside the inclusion fence (fence-aware path planning).	M	S	SIM	NAV-007, V-01
HLM-D20	The heading estimate shall fall back to GNSS-velocity yaw when the compass disagrees above 0.3 m/s.	S	S	SIM	NAV-008, V-16

5.5 Fence

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D21	The helm shall hold one inclusion polygon (≤ 70 vertices) and ≤ 10 exclusion polygons or circles, plus a 100 m circle backstop about home at Milton.	M	S	SIM	FEN-001, OPS-006
HLM-D22	The fence shall be enforced in every armed mode used.	M	S	SIM	FEN-004, V-02
HLM-D23	A breach shall trigger RTL within 1 s.	M	S	SIM	FEN-005
HLM-D24	Persistent breach (> 30 s or > 10 m outside) shall stop the motors. Native mechanism to be confirmed (V-14). Interim: Mission Control and the MCP both command HOLD.	M	S	SIM	FEN-006, V-14
HLM-D25	Fence changes while armed shall be refused by every sender (IF-14, IF-04 filter) and by the helm if supported (V-17).	M	S	SIM	FEN-007, V-17

5.6 Failsafes

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D26	Battery failsafe: RTL at 35% remaining; at 15%, continue RTL, alarm and reduce speed (V-15).	M	S	SIM	FS-001, V-15
HLM-D27	GCS failsafe on system-255 heartbeats: 2 s timeout, action HOLD, continue in AUTO. The later RTL steps are done by MCP B4 (MCP-D12).	M	S	SIM	FS-002, FS-003, V-03, V-06
HLM-D28	EKF/position failsafe $\rightarrow$ HOLD (motors stop). Recovery RTL is done by MCP B6 (MCP-D15).	M	S	SIM	FS-004
HLM-D29	Crash/stuck check $\rightarrow$ HOLD, raising an event the MCP can act on.	M	S	POOL	FS-005, V-05
HLM-D30	No companion-computer failsafe shall be configured: the helm completes missions without the MCP.	M	S	SIM	FS-007, SAF-002
HLM-D31	When failsafes combine, motors-stopped outranks RTL, which outranks continue. Demonstrated in SITL combinations.	M	S	SIM	FS-011
HLM-D32	Every failsafe shall be logged and reported by STATUSTEXT.	M	S	SIM	FS-012

5.7 Independence, identity and logging

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D33	All fence and failsafe functions shall work with the MCP, Mission Control and the link all absent.	M	S	SIM	SAF-002
HLM-D34	Only system 255 is the GCS. Parameter writes are accepted only while disarmed through IF-14. Boat services are filtered (IF-04).	M	T	SIM	SAF-003, IF-02, IF-04
HLM-D35	The GNSS/compass shall be mast-mounted ≥ 150 mm from power wiring, with motor interference $\leq 30\%$ after a motor-interference calibration.	M	T	BENCH	MEC-013
HLM-D36	The helm shall log position, attitude, mode, commands, battery and failsafes at ≥ 5 Hz.	M	T	BENCH	LOG-001
HLM-D37	A spare UART shall be kept for an optional ExpressLRS receiver, with RC failsafe = RTL if fitted.	C	I	BENCH	COM-007
HLM-D38	The beacon output shall be configured per IF-09.	S	T	BENCH	IF-09, REC-004

6. Controlled parameter baseline

The baseline configuration that implements the requirements above. **The authoritative list is `params/boaty-mk1.parm` in the repository.** Names and values are confirmed against the flashed firmware version in SITL before the file is frozen (HLM-D01). 'Confirm' shows where each is proven.

Function	Parameter(s)	Baseline	Satisfies	Confirm
Vehicle	FRAME_CLASS	Boat	MOD-001	V-01
Motor outputs	SERVO1_FUNCTION / SERVO2_FUNCTION	ThrottleLeft (73) / ThrottleRight (74)	NAV-001, IF-05	Bench
Output protocol	MOT_PWM_TYPE	DShot300	IF-05	TBC-05
Throttle cap	MOT_THR_MAX	Tuned: full throttle ≤ 1.5 m/s	NAV-003	Pool
Cruise	CRUISE_SPEED / CRUISE_THROTTLE	1.0 m/s / measured	NAV-002	Pool
Waypoints	WP_SPEED / WP_RADIUS	1.0 m/s / 3 m	NAV-004	SITL
RTL	RTL_SPEED	0 (= WP_SPEED)	NAV-007	SITL
Loiter	LOIT_RADIUS	2 m	NAV-006	Lake
Path planning	OA_TYPE	Dijkstra (fence-aware)	NAV-007	V-01
Identity	SYSID_THISMAV / SYSID_MYGCS	1 / 255	IF-02	V-06
Companion port	SERIAL2_PROTOCOL / SERIAL2_BAUD	2 (MAVLink 2) / 115	IF-04	TBC-02
Fence	FENCE_ENABLE / FENCE_TYPE / FENCE_ACTION	1 / polygon + circle / RTL	FEN-001, FEN-005	V-02
Fence backstop	FENCE_RADIUS / FENCE_MARGIN	100 m / 3 m	OPS-006	SITL
Battery	BATT_MONITOR / BATT_CAPACITY	Analogue V+I / measured (≈ 3000 mAh)	PWR-006	Bench
Battery failsafe	BATT_LOW_MAH / BATT_CRT_MAH	35% / 15% of capacity	FS-001	SITL
Battery actions	BATT_FS_LOW_ACT / BATT_FS_CRT_ACT	RTL / RTL	FS-001	V-15
Arming capacity	BATT_ARM_MAH	80% of capacity	PRE-003	SITL
Motor rail	BATT2_MONITOR / BATT2_ARM_VOLT	Analogue V / 9.0 V	MOD-003, PRE-007	V-13
GCS failsafe	FS_GCS_ENABLE / FS_TIMEOUT / FS_ACTION	2 (continue in AUTO) / 2 s / Hold	FS-002, FS-003	V-03
EKF failsafe	FS_EKF_ACTION	Hold	FS-004	SITL
Stuck	FS_CRASH_CHECK	Hold	FS-005	V-05
GNSS	GPS_TYPE / GPS_HDOP_GOOD	Auto / 150	PRE-001	Bench
Arming	ARMING_CHECKS	All	MOD-004	SITL
Heading	COMPASS_ORIENT (+ EKF3 GSF yaw)	Per module / default on	NAV-008	TBC-09, V-16
Beacon	NTF_LED_TYPES / SERVO3_FUNCTION	NeoPixel / NeoPixel1	REC-004, IF-09	Bench
Logging	LOG_BACKEND_TYPE / LOG_BITMASK	File / default + ≥ 5 Hz	LOG-001	Bench
Mission end	MIS_DONE_BEHAVE	Hold (backstop)	MOD-005	V-04

6.1 Where each failsafe step lives

Requirement	Helm (native)	Boat Python (MCP)	Mission Control
FS-001	RTL at 35%; RTL + alarm at 15%	-	Announce; reduce speed if V-15 shows no native way
FS-002	HOLD at 2 s	RTL at 10 s (B4)	Show 'link lost'
FS-003	Continue in AUTO	RTL at 60 s (B4)	-
FS-004	HOLD (motors stop)	RTL after 10 s healthy (B6)	Announce
FS-005/006	HOLD on stuck	≤ 3 astern bursts (B5)	Announce
FEN-005/006	RTL on breach; persistent → stop (V-14)	HOLD if persistent (interim)	HOLD if persistent (interim)

Rule: nothing outside the helm is the last line of defence. Boat Python and Mission Control only add later, safer steps.

7. Design constraints and budgets

- ADD DD-02: F405-class, subject to V-01
- ADD DD-13: no Lua; Python on the MCP covers Lua-type behaviours
- No project-written code on the FC (SAF-001)

Budget	Allocation
Mass allocation	≤ 40 g (estimate 32 g)
Cost allocation	£39 (FC £25, GNSS £14)
Power	≈ 0.8 W including GNSS

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	1	-	1	-	20	22
BENCH	8	-	4	-	-	12
POOL	2	-	-	-	1	3
LAKE	1	-	-	-	-	1

Ref	Open item
V-01	Firmware feature check for the chosen board, incl. path planning around fences
V-02 ... V-06	Mode, failsafe and identity behaviours in SITL
V-14	Native mechanism for FEN-006 (persistent breach → motors stop)
V-15	Reduced RTL speed on critical battery (FS-001)
V-16	GNSS-velocity yaw fallback behaviour (NAV-008)
V-17	Whether the helm itself refuses fence changes while armed
HLM-OI-1 / CR-02	PRE-008 asks for ≥ 1 GB free on the helm. Many F405 boards log to 16 MB dataflash. Either choose a board with microSD, or relax PRE-008 for the helm (owner decision)

PROJECT BOATY

Subsystem Specification

Mission computer

Take and geotag photos, relay MAVLink to the bank, and run the small boat-side watchdogs, without ever being needed for safety.

Document	BOATY-SSS-MCP
Issue	Issue A (for review)
Date	28 September 2026
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Parents	SRS Issue D, ADD Issue C, ICD Issue B
Content	12 allocated SRS requirements → 21 subsystem requirements (13 M, 6 S, 2 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Take and geotag photos, relay MAVLink to the bank, and run the small boat-side watchdogs, without ever being needed for safety.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Raspberry Pi Zero 2W, microSD, heatsink • 5 MP camera • Moisture sensor • OS image and services B1-B6	• Its 5 V supply (PWR) • Camera hood and mount (HUL) • Helm behaviour (HLM)

1.3 Breakdown

Item	Description
MCP-1 Computer	Pi Zero 2W, 32 GB A1/U3 microSD, heatsink
MCP-2 Camera	OV5647-class 5 MP, $\leq 62^\circ$ HFOV
MCP-3 Sensors	Moisture traces at the box low point
MCP-4 Software	B1 router, B2 camera, B3 photo server, B4 link watchdog, B5 weed-shedding, B6 health

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue C	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue B	Interface definitions (all listed in section 3 apply in full)

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-01 Boat-bank radio link	Mission Control	Conforms	\$4
IF-03 Photo & health service	Mission Control	Owner	\$4
IF-04 Companion link	Helm	Owner	\$4
IF-08 Mission computer power	Power	Conforms	\$6

4. Parent requirements

Primary (12): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
FS-006	S	Weed-shedding: after a stuck detection the boat shall try up to 3 short reverse bursts, then resume its previous mode...	MCP-D18
FS-010	S	Water ingress: a moisture sensor in the electronics box shall trigger RTL and an alarm.	MCP-D03, MCP-D15
CAM-001	M	The boat shall carry a forward-looking camera that captures still images of at least 2 MP (M) and at least 5 MP (S).	MCP-D01
CAM-002	M	Capture modes shall be: interval (every 2 to 30 s), at photo points, and on command.	MCP-D10
CAM-003	M	Every photo shall be tagged with UTC time, position, heading and mission ID.	MCP-D11
CAM-004	M	Local storage shall hold at least 500 photos. Full storage shall not affect navigation or safety.	MCP-D06, MCP-D13
CAM-005	S	A duck (about 0.3 m long) at 5 m shall span at least 40 pixels in the image.	MCP-D02
CAM-007	S	After a mission, photos shall transfer to Mission Control automatically when in range, with 200 photos taking 5 min or...	MCP-D14, MCP-D16
CAM-008	C	Mission Control may show a low-rate live view from the camera.	MCP-D17
DET-003	C	An upgraded mission computer may detect waterfowl on board in real time and trigger 'pause and look' (hold and take a...	MCP-D20
DET-004	M	If DET-003 is implemented, 'pause and look' shall hold position and shall never steer towards the detected animal.	MCP-D21
PWR-010	M	Electronics in the sealed box shall run for 60 min at 30 °C ambient in direct sun without fault or throttling.	MCP-D04

Supporting (14): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

PRE-008, FS-003, FS-007, CAM-006, COM-001, COM-002, COM-003, COM-004, COM-006, PWR-005, SWE-004, SWE-008, LOG-004, ENV-005

5. Subsystem requirements

Requirements are numbered MCP-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Hardware

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D01	The mission computer shall be a Pi Zero 2W with a 32 GB A1/U3 microSD and a 5 MP camera.	M	I	BENCH	CAM-001, DD-03
MCP-D02	Camera horizontal field of view shall be $\leq 62^\circ$, giving ≥ 120 px across a 0.3 m duck at 5 m (at 2592 px width).	S	A	BENCH	CAM-005
MCP-D03	A moisture sensor (exposed traces at the box's lowest point) shall be read by GPIO.	S	T	BENCH	FS-010
MCP-D04	The SoC shall have a heatsink, and 60 min at 30 °C in the sealed box in sun shall give no throttling (throttled flag = 0).	M	T	BENCH	PWR-010, V-10
MCP-D05	Average power ≤ 2.0 W, peak ≤ 3.0 W.	S	T	BENCH	PWR-011

5.2 Platform software

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D06	The OS shall be Raspberry Pi OS Lite 64-bit with a read-only root filesystem and a separate data partition.	M	T	BENCH	CAM-004, IF-08
MCP-D07	Services B1-B6 shall be systemd units that restart automatically and are running ≤ 45 s after power-on.	M	T	BENCH	IF-04, IF-03
MCP-D08	The clock shall be set from helm GNSS time at start (no internet on the boat).	S	T	BENCH	SWE-008
MCP-D09	SSH shall be key-only, and the firewall per IF-01.	M	I	BENCH	COM-006, IF-01

5.3 Camera and photos (B2, B3)

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D10	B2 shall capture on interval (2-30 s during AUTO), at photo points (burst_n within 0.5 s of arrival) and on command.	M	T	SIM	CAM-002, IF-03
MCP-D11	Each photo shall carry UTC time, lat/lon/heading (≤ 0.5 s old, else null) and mission ID, in EXIF and the index.	M	T	SIM	CAM-003
MCP-D12	B4 (link watchdog): in AUTO with no system-255 heartbeat for 60 s $\rightarrow$ RTL. In HOLD after a MANUAL link loss, at 10 s $\rightarrow$ RTL.	M	S	SIM	FS-003, FS-002
MCP-D13	≥ 5 GB shall be reserved for photos (≥ 500 at ≤ 2.5 MB). Capture stops below 500 MB free. Navigation is unaffected.	M	T	BENCH	CAM-004
MCP-D14	B3 shall implement IF-03, with thumbnails made at capture time.	M	T	BENCH	IF-03, CAM-007
MCP-D15	B6 (health): moisture $\rightarrow$ RTL within 2 s plus STATUSTEXT. After a position-loss HOLD, RTL once the EKF has been healthy for 10 s.	M	S	SIM	FS-010, FS-004
MCP-D16	Photo sync throughput ≥ 11 Mbit/s with the boat ≤ 10 m from the bank (200 photos ≤ 5 min).	S	T	POOL	CAM-007
MCP-D17	Optional live view: 320 $\times$ 240 MJPEG at 2 fps when enabled, paused during sync.	C	D	POOL	CAM-008

5.4 Weed-shedding and command filter (B5)

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D18	B5 shall act only on a helm stuck event. Up to 3 GUIDED astern bursts (≤ 0.5 m/s, ≤ 2 s, targets at 10 Hz), then resume the previous mode if ground speed > 0.3 m/s is regained, else HOLD + alarm.	S	S	SIM	FS-006, V-11
MCP-D19	All services shall send MAVLink only through a shared client that blocks forbidden messages (IF-04 list).	M	T	SIM	SAF-003, IF-04

5.5 Future: on-board duck spotting

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D20	An upgrade may run an on-device bird detector at ≥ 2 fps that requests 'pause and look'.	C	T	POOL	DET-003
MCP-D21	'Pause and look' shall only request LOITER (≤ 20 s) and then resume. It shall never steer towards the detection. Applies only if MCP-D20 is implemented.	M	S	SIM	DET-004

6. Timing

Event	Response	Budget
MISSION_ITEM_REACHED (photo point)	First photo of the burst	≤ 0.5 s
Interval tick	Photo captured	± 0.2 s
Moisture detected	RTL requested	≤ 2 s
No system-255 heartbeat in AUTO	RTL requested	$60\text{ s} \pm 1\text{ s}$
Helm stuck event	First astern burst	≤ 1 s
Power-on	All services running	≤ 45 s
POST /v1/shutdown	Filesystems synced, halt	≤ 10 s

7. Design constraints and budgets

- ADD DD-03 / T4: Pi Zero 2W (Python)
- ADD section 4.3 rule: may request only safer states; B5 bounded GUIDED
- SWE-001: Python services

Budget	Allocation
Mass allocation	≤ 35 g (estimate 30 g)
Cost allocation	£27 (Pi £15, camera £8, SD £4)
Power	≤ 2.0 W average, ≤ 3.0 W peak

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	3	-	-	-	4	7
BENCH	8	-	2	1	-	11
POOL	2	1	-	-	-	3
LAKE	-	-	-	-	-	0

Ref	Open item
V-10	Thermal in the sealed box
V-11	Helm stops if B5 dies mid-burst

Subsystem Specification

Mission Control

Be the only place people interact with Boaty: turn words into safe, approved missions, and show, say and record what the boat is doing.

Document	BOATY-SSS-MCN
Issue	Issue A (for review)
Date	28 September 2026
Status	For review by the project owner
Parents	SRS Issue D, ADD Issue C, ICD Issue B
Content	65 allocated SRS requirements → 52 subsystem requirements (35 M, 16 S, 1 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Be the only place people interact with Boaty: turn words into safe, approved missions, and show, say and record what the boat is doing.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Raspberry Pi 5, case and panel (4 buttons, key switch, LEDs) • USB Wi-Fi adapter and antenna (pole kit if needed) • USB mic and speaker • Application components C1-C10 • Site files, logs, captain's logs	• The phone/tablet (display, internet) • The Claude API • Helm behaviour (HLM)

1.3 Breakdown

Item	Description
MCN-1 Hardware	Pi 5, cooler, case, panel, radio, audio
MCN-2 Session & UI	C1 session manager, C2 panel I/O, C3 web UI, C4 voice
MCN-3 Mission chain	C5 planner, C6 validator, C7 helm interface
MCN-4 Records	C8 photos & log, C9 site store, C10 logger

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue C	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue B	Interface definitions (all listed in section 3 apply in full)

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-01 Boat-bank radio link	Mission computer	Owner	\$4
IF-02 Command & telemetry	Helm	Owner	\$4
IF-03 Photo & health service	Mission computer	Conforms	\$4
IF-10 Internet uplink	Phone	Owner	\$4
IF-11 Claude API	Claude API	Owner	\$4
IF-12 Crew & operator panel	Operator & crew	Owner	\$7
IF-13 Mission format	internal	Owner	\$5
IF-14 Helm API	internal	Owner	\$5
IF-15 Site data	Simulation & test	Owner	\$5
IF-21 Simulation link	Simulation & test	Conforms	\$5

4. Parent requirements

Primary (65): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
MOD-006	M	While armed, the operator shall be able to command HOLD, RTL, MANUAL or STOP from Mission Control at any time,...	MCN-D11
MOD-007	S	Every mode change shall be shown and announced at Mission Control within 2 s, whatever caused it.	MCN-D12
PRE-005	M	For AUTO: a mission that has passed validation and operator approval shall be loaded, and read back from the helm to...	MCN-D15, MCN-D43
PRE-006	M	The pre-launch checklist (OPS-001) shall be complete and signed off at Mission Control for this session.	MCN-D13
FEN-002	M	Fences shall be drawn on a map at Mission Control, stored per site with a name, and version-controlled.	MCN-D19
FEN-003	S	Mission Control shall show a 5 m guide inside the drawn inclusion fence, and warn if any home or waypoint falls within...	MCN-D19
FS-009	M	STOP: pressing STOP at Mission Control shall stop both motors within 1 s, whatever the mode.	MCN-D10
SAF-004	M	Output from the language model shall never be sent to the helm as a command. Only mission objects that have passed...	MCN-D38
SAF-007	S	Safety parameters (fence, failsafe thresholds, speed limits) shall be version-controlled. The values loaded on the...	MCN-D45
MIS-001	M	The system shall support these mission elements: waypoint, area survey (lawnmower pattern over a polygon, with lane...	MCN-D34
MIS-002	M	Every mission shall end with return home.	MCN-D34
MIS-003	M	Planned mission duration shall not exceed 20 min (configurable up to 30 min). Planned energy shall not exceed 50% of...	MCN-D35
MIS-004	M	Missions shall be represented in a documented JSON schema with a version number.	MCN-D36
MIS-005	M	Offline templates shall be available without internet, at least: 'Explore the bay', 'Duck patrol' (survey with photo...	MCN-D37
MIS-006	S	An adult shall be able to create and edit missions on the map at Mission Control.	MCN-D20
NLI-001	M	Mission Control shall accept instructions as typed text.	MCN-D26
NLI-002	S	Mission Control shall accept spoken instructions through a microphone, using push-to-talk on the TALK button.	MCN-D25
NLI-003	M	Instructions shall be turned into a mission through the Claude API. The request shall include the site context: fence,...	MCN-D28
NLI-004	M	The language model's reply shall be constrained to the mission schema (structured output or tool use). A malformed...	MCN-D29
NLI-005	M	The proposed mission shall be shown on the map with a one-sentence plain-language summary, which shall also be spoken...	MCN-D30
NLI-006	M	Requests the boat must not or cannot do (for example leaving the fence, chasing animals, going to another lake) shall...	MCN-D31
NLI-007	S	With mobile data available, a plan shall be displayed within 20 s of the instruction.	MCN-D32
NLI-008	M	The API key shall be stored only on Mission Control and shall never be committed to the repository.	MCN-D33
VAL-001	M	Every mission, from any source, shall pass the validator before upload. There shall be no other path for uploading...	MCN-D38
VAL-002	M	Every waypoint, and every leg sampled at 1 m intervals or finer, shall be inside the inclusion fence by at least 3 m...	MCN-D39

SRS	Pri	Requirement (abridged)	Met by
VAL-003	M	The mission shall meet the duration and energy limits in MIS-003.	MCN-D39
VAL-004	M	The mission shall start within 10 m of home and end with return home.	MCN-D39
VAL-005	M	Commanded speeds shall be within NAV-003. Each hold shall be 60 s or less. The photo count shall fit in the free...	MCN-D39
VAL-006	M	The validator shall be pure Python with no network access and no randomness. It shall have a unit-test suite that...	MCN-D40
VAL-007	M	Each rejection shall be reported in plain language, naming the rule broken and where on the map.	MCN-D41
VAL-008	M	After validation, an adult shall approve the mission at Mission Control. GO shall stay disabled until then.	MCN-D08, MCN-D42
VAL-009	M	Any change to an approved mission shall void the approval.	MCN-D42
VAL-010	M	After upload, the mission shall be read back from the helm and compared with the approved mission by checksum before...	MCN-D43
DET-001	S	After a mission, the system shall identify photos containing waterfowl, with at least 80% precision on a test set of...	MCN-D48
DET-002	S	The system shall produce a captain's log: up to 6 best photos, a map of the route, distance, duration, ducks seen, and...	MCN-D49
MC-001	M	Mission Control shall run on the owned Raspberry Pi 5 and operate for at least 2.5 h from a USB-C power bank.	MCN-D01
MC-002	M	It shall have four physical push buttons, each at least 30 mm across, in distinct colours with icons: GO (green), COME...	MCN-D02
MC-003	M	STOP and COME HOME shall always be active while armed. GO shall be active only when a validated, approved and verified...	MCN-D08
MC-004	S	GO shall require a 1 s press-and-hold, to prevent accidental starts.	MCN-D09
MC-005	M	The display shall show, updated at least once a second: a map with the boat's position, heading and track; fence,...	MCN-D16
MC-006	S	The display shall be readable in daylight. A tablet or phone used as the display is acceptable.	MCN-D17
MC-007	S	Mission Control shall give spoken announcements of mode changes, failsafes and mission events in child-friendly words.	MCN-D27
MC-008	M	Adult-only functions (arming, fence editing, parameter changes, mission approval) shall be protected by a PIN or key...	MCN-D14
MC-009	M	The last known position and track shall stay displayed after link loss, in a 'find my boat' view.	MCN-D18
MC-010	M	Mission Control shall host the pre-launch checklist (OPS-001) and record who signed it off, and when.	MCN-D13
MC-011	M	Everything except natural-language planning and cloud photo analysis shall work without internet access.	MCN-D07
MC-012	S	An adult shall be able to drive the boat in MANUAL from Mission Control, using an on-screen joystick or a gamepad.	MCN-D22
MC-013	S	The Mission Control case shall resist splashes (IPX4 intent).	MCN-D01
MC-015	S	Mission Control shall have a photo-review mode the crew can use: large thumbnails, stepped through with the buttons,...	MCN-D23
MC-014	S	QGroundControl or MAVProxy shall be usable as a backup ground station.	MCN-D24
COM-001	M	The boat-to-bank link shall use licence-exempt 2.4 GHz Wi-Fi within UK power limits.	MCN-D03
COM-002	M	Link range shall be at least 100 m line of sight over water, with the boat antenna at least 150 mm above the water.	MCN-D03

SRS	Pri	Requirement (abridged)	Met by
COM-003	M	Within range: telemetry at 1 Hz or faster, and command latency of 1 s or less (95th percentile).	MCN-D46
COM-004	S	Link quality shall be shown at Mission Control and logged.	MCN-D21
COM-005	M	Mission Control shall reach the internet through a phone hotspot. Losing internet shall not affect the boat link.	MCN-D04
COM-006	M	The link shall be secured with WPA2 or better and non-default credentials. The boat shall accept commands only from...	MCN-D03
SWE-002	M	A Python helm interface shall provide: arm, disarm, set mode, upload mission, read mission back, start, hold, return...	MCN-D44
SWE-003	M	Mission-level code shall run unchanged against any helm implementation, selected by configuration: first ArduPilot...	MCN-D44
SWE-008	S	All components shall timestamp in UTC, taken from GNSS where available.	MCN-D06
SWE-009	S	Mission Control software shall start automatically within 60 s of power-on.	MCN-D05
LOG-002	M	Mission Control shall log: all commands, sign-offs and approvals; each mission as approved (JSON); language model...	MCN-D50
LOG-003	M	The logs shall be enough to replay any trip on a map, including every failsafe event.	MCN-D51
LOG-004	M	Photos shall be stored locally and never published automatically. Photos go to cloud services only if an adult enables...	MCN-D47
LOG-005	C	Logs and photos may be backed up at home and kept for at least 12 months.	MCN-D52
CHD-005	M	The crew controls (GO, COME HOME, STOP, TALK) shall be usable without reading, by colour and icon alone.	MCN-D02

Supporting (22): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

MOD-001, MOD-003, MOD-004, MOD-005, PRE-001, PRE-002, PRE-004, PRE-008, FEN-001, FEN-004, FEN-005, FEN-006, FEN-007, FS-012, SAF-001, SAF-002, SAF-003, CAM-007, CAM-008, SWE-004, OPS-001, CON-002

5. Subsystem requirements

Requirements are numbered MCN-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Hardware and platform

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D01	Mission Control shall be a Pi 5 (≥ 4 GB) with an active cooler in a printed splash-resistant case with drip edges. ≥ 2.5 h on a 20,000 mAh USB-C PD bank.	M	T	BENCH	MC-001, MC-013
MCN-D02	The panel shall have TALK, GO, COME HOME and STOP buttons (≥ 30 mm, colour + icon, LEDs) and an adult key switch, per IF-12.	M	I	BENCH	MC-002, CHD-005, IF-12
MCN-D03	The radio shall be a USB adapter with an RP-SMA antenna, acting as AP per IF-01 (country GB, per-site channel, WPA2+, non-default passphrase).	M	T	LAKE	COM-001, COM-002, COM-006
MCN-D04	Internet shall come only via the phone USB tether per IF-10. The boat network has no route out.	M	T	BENCH	COM-005, IF-10
MCN-D05	The application shall start automatically ≤ 60 s after power-on.	S	T	BENCH	SWE-009
MCN-D06	The clock shall be disciplined from helm GNSS time when there's no internet. All logs are UTC.	S	T	BENCH	SWE-008
MCN-D07	Everything except natural-language planning and cloud photo analysis shall work offline, including cached map tiles per site.	M	T	BENCH	MC-011

5.2 C1 Session manager and C2 panel

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D08	The session state machine (ICD Figure 4) shall gate every button. GO works only in ARMED with a verified mission. COME HOME and STOP work in every armed state.	M	T	SIM	MC-003, VAL-008
MCN-D09	GO shall need a 1.0 ± 0.1 s press-and-hold.	S	T	BENCH	MC-004
MCN-D10	STOP shall call helm.stop() ≤ 50 ms after the button edge. End-to-end ≤ 1 s (IF-02 budget).	M	T	POOL	FS-009
MCN-D11	An adult shall be able to command HOLD, RTL, MANUAL or STOP at any time while armed, overriding AUTO.	M	T	SIM	MOD-006
MCN-D12	Mode changes shall be shown and spoken ≤ 2 s after the helm reports them. Auto-disarm after 60 s at home.	S	T	SIM	MOD-007, MOD-008
MCN-D13	The checklist shall hold every OPS-001 item and record who signed it off, and when. Arming is blocked until it's done.	M	D	SIM	PRE-006, MC-010
MCN-D14	Arming, approval, fence editing, parameter changes, manual drive and the cloud toggle shall need the adult key switch (PIN fallback).	M	T	SIM	MC-008
MCN-D15	Before arming, Mission Control shall also check ≥ 8 satellites, a verified mission and ≥ 1 GB free locally.	M	T	SIM	PRE-005, PRE-001, PRE-008

5.3 C3 Web UI

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D16	The map view shall show boat, heading, track, fence, exclusions, home, mission, battery, link, mode and time left, at ≥ 1 Hz.	M	D	SIM	MC-005
MCN-D17	A high-contrast daylight theme (≥ 16 px text) shall work on phone and tablet.	S	D	LAKE	MC-006

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D18	The 'find my boat' view shall keep the last position, time and track after link loss.	M	T	SIM	MC-009
MCN-D19	The fence editor shall show the 5 m guide, warn on violations, save to the site file and commit it to git.	M	D	SIM	FEN-002, FEN-003, IF-15
MCN-D20	An adult map mission editor shall route every edit through the validator.	S	D	SIM	MIS-006
MCN-D21	Link quality (RSSI both ends) shall be shown and logged at 1 Hz.	S	D	POOL	COM-004
MCN-D22	Manual drive from an on-screen joystick or gamepad: 10 Hz MANUAL_CONTROL with a dead-man (release → zero).	S	D	POOL	MC-012
MCN-D23	Crew photo review: large thumbnails, GO = next, COME HOME = back to start, ducks highlighted.	S	D	SIM	MC-015
MCN-D24	With the application paused, QGroundControl shall be able to take over UDP 14550 as a backup ground station.	S	D	SIM	MC-014

5.4 C4 Voice

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D25	TALK held → record (≤ 15 s) → on-device STT → transcript shown to the adult. Audio is never stored or sent.	S	T	BENCH	NLI-002, V-07
MCN-D26	A typed instruction box shall be available on the web UI.	M	D	SIM	NLI-001
MCN-D27	Spoken announcements shall use the IF-12 phrase set via on-device TTS.	S	D	SIM	MC-007

5.5 C5 Planner

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D28	Requests shall follow IF-11 and include site context (names, sizes, battery, limits). No coordinates.	M	T	SIM	NLI-003, IF-11
MCN-D29	Replies shall be validated against Intent v1. On failure, retry ≤ 2 times, then offer templates.	M	T	SIM	NLI-004
MCN-D30	The plan shall be drawn on the map, and its summary_for_child shown and spoken.	M	D	SIM	NLI-005
MCN-D31	Declines (off-site names, chasing animals, leaving the fence) shall come from the system prompt rules and planner checks, worded for a child, with an alternative offered.	M	T	SIM	NLI-006
MCN-D32	Plan display ≤ 20 s at the 95th percentile on 4G.	S	T	LAKE	NLI-007
MCN-D33	The API key shall live in a 0600 file outside the repository. CI runs a secret scan.	M	I	SIM	NLI-008
MCN-D34	The geometry generator shall make survey lanes (light 20 m, medium 12 m, thorough 8 m spacing), photo stops $\geq$ keep_out_m from landmarks, and a final RTL.	M	T	SIM	MIS-001, MIS-002
MCN-D35	The duration and energy estimate shall be conservative ($\times 1.2$) and enforce 20 min (max 30) and 50% of usable energy.	M	T	SIM	MIS-003
MCN-D36	Intent, Mission and ValidationResult shall be versioned pydantic models per IF-13.	M	I	SIM	MIS-004, IF-13
MCN-D37	At least three offline templates, parameterised by site areas.	M	D	SIM	MIS-005

5.6 C6 Validator and C7 helm interface

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D38	helm.upload_mission shall accept only a ValidatedMission type that only the validator can create (type-level single path).	M	T	SIM	VAL-001, SAF-004

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D39	The validator shall implement VAL-002 to VAL-005 in local ENU metres, sampling legs every ≤ 1 m.	M	T	SIM	VAL-002, VAL-003, VAL-004, VAL-005
MCN-D40	The validator shall do no I/O and be deterministic, with $\geq 95\%$ branch coverage and ≥ 50 adversarial cases.	M	T	SIM	VAL-006
MCN-D41	Violations shall be reported with rule, adult and child messages, and a map location.	M	D	SIM	VAL-007
MCN-D42	Approval shall bind to the mission checksum. Any change voids it.	M	T	SIM	VAL-008, VAL-009
MCN-D43	GO shall be enabled only after read-back checksum equality (IF-02).	M	T	SIM	VAL-010, PRE-005
MCN-D44	C7 shall implement the IF-14 Helm protocol (ArduPilotHelm, pymavlink), selected by configuration.	M	T	SIM	SWE-002, SWE-003, IF-14
MCN-D45	On connect, C7 shall compare helm parameters with the committed file. A mismatch blocks arming and shows the difference.	S	T	SIM	SAF-007
MCN-D46	Telemetry ≥ 1 Hz and command latency ≤ 1 s (P95) shall be measured and logged every session.	M	T	POOL	COM-003

5.7 C8 Photos & log, C10 logger

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D47	Photos shall be synced per IF-03 after each mission and sha256-verified before ack. Kept locally, never published automatically. Cloud analysis off by default per session.	M	T	BENCH	LOG-004, CAM-007
MCN-D48	The waterfowl finder shall reach $\geq 80\%$ precision on the lake test set (cloud when enabled, else on-device heuristic).	S	T	LAKE	DET-001
MCN-D49	The captain's log shall have ≤ 6 photos, route map, distance, duration, ducks and a child-friendly story, saved as HTML/PDF.	S	D	LAKE	DET-002
MCN-D50	The session log shall be structured JSON lines: commands, approvals, missions, LLM I/O, validator results, events, RSSI.	M	I	SIM	LOG-002
MCN-D51	After each session the helm log shall be downloaded. A replay tool shows the trip and every failsafe on a map.	M	D	SIM	LOG-003
MCN-D52	Sessions may sync to a home computer when on the home network, kept ≥ 12 months.	C	D	SIM	LOG-005

6. Component map

Mission Control's application components (ADD section 4.3) and the derived requirements each implements. Button behaviour by state is defined in ICD IF-12.

ID	Component	Responsibility	Derived requirements
C1/C2	Session manager, panel I/O	States, gating, checklist, adult key	MCN-D08 ... D15
C3	Web UI	Map, editors, photo review, manual	MCN-D16 ... D24
C4	Voice	TALK, STT, TTS	MCN-D25 ... D27
C5	Planner	Claude intent, geometry, templates	MCN-D28 ... D37
C6/C7	Validator, helm interface	Single mission path, read-back, parameters	MCN-D38 ... D46
C8/C10	Photos & log, logger	Sync, finder, captain's log, replay	MCN-D47 ... D52
C9	Site store	GeoJSON sites (IF-15)	MCN-D19

7. Design constraints and budgets

- ADD AR-2: intelligence on the bank
- ADD DD-04: LLM returns intent; Python makes geometry
- ADD DD-06/07/08: Pi 5 AP, on-device speech, web UI

Budget	Allocation
Mass	Not constrained (bank)
Cost allocation	£29 (buttons/key £9, mic/speaker £8, adapter £12) + £8 pole kit if needed
Power	≈ 6.5 W average from a USB-C PD bank

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	18	13	3	-	-	34
BENCH	8	-	1	-	-	9
POOL	2	2	-	-	-	4
LAKE	3	2	-	-	-	5

Ref	Open item
TBC-10	Closed: TALK button (ICD Issue B)
V-07	On-device speech latency and child accuracy
V-08	Link range with and without the pole

PROJECT BOATY

Subsystem Specification Recovery & signalling

Make the boat easy to see, easy to find and easy to retrieve with no power, no software and no radio.

Document	BOATY-SSS-REC
Issue	Issue A (for review)
Date	28 September 2026
Status	For review by the project owner
Parents	SRS Issue D, ADD Issue C, ICD Issue B
Content	6 allocated SRS requirements → 8 subsystem requirements (6 M, 1 S, 1 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Make the boat easy to see, easy to find and easy to retrieve with no power, no software and no radio.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Hi-vis finish • Flag • Recovery hoop • LED beacon ring (hardware) • 'If found' labels • Optional tracker pocket • Bank recovery kit	• Beacon drive and patterns (HLM) • Mast (HUL)

1.3 Breakdown

Item	Description
REC-1 Visibility	Fluorescent parts/tape, flag
REC-2 Retrieval	Hoop, recovery kit
REC-3 Identification	Labels, tracker pocket
REC-4 Beacon	WS2812 ring at the mast head

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue C	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue B	Interface definitions (all listed in section 3 apply in full)

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-09 Beacon drive	Helm	Conforms	§6

4. Parent requirements

Primary (6): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
REC-002	M	At least 50% of the visible top surface shall be fluorescent orange or yellow.	REC-D01
REC-003	M	A flag at least 100 × 60 mm shall fly with its top at least 300 mm above the waterline.	REC-D02
REC-004	S	An LED beacon shall be visible from 100 m in overcast daylight, with flash patterns showing armed, failsafe and...	REC-D03
REC-005	M	A recovery hoop of at least 60 mm internal diameter shall be fitted to the mast. It shall withstand a 50 N pull, and...	REC-D06
REC-006	M	The hull shall carry an 'If found' label with a contact phone number.	REC-D04
REC-007	C	A Bluetooth item tracker may be carried in the hull.	REC-D05

5. Subsystem requirements

Requirements are numbered REC-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Visibility and identification

ID	Requirement	Pri	Ver	Stage	Trace
REC-D01	At least 50% of the visible top area shall be fluorescent orange or yellow, measured on a top-down photo.	M	T	BENCH	REC-002
REC-D02	A fluorescent ripstop flag $\geq 120 \times 80$ mm shall fly with its top ≥ 320 mm above the waterline, below the GNSS.	M	I	BENCH	REC-003, IF-19
REC-D03	The beacon ring shall be visible at 100 m in overcast daylight and show the helm's state patterns (IF-09).	S	T	LAKE	REC-004, IF-09
REC-D04	Both hulls shall carry a waterproof 'IF FOUND' label with the owner's phone number.	M	I	BENCH	REC-006
REC-D05	An optional sealed pocket in a mid segment shall take a ≤ 35 mm tracker tag.	C	I	BENCH	REC-007

5.2 Retrieval

ID	Requirement	Pri	Ver	Stage	Trace
REC-D06	A printed hoop (≥ 60 mm internal) integrated with the mast collar shall take a 50 N pull. The boat tows by it without capsizing.	M	T	POOL	REC-005
REC-D07	The bank kit shall contain a ≥ 3 m telescopic pole with net and a casting rod with a weighted float line, packed with the boat.	M	I	LAKE	OPS-008
REC-D08	The flag staff shall be capped, and hoop and flag edges rounded (≥ 0.6 mm).	M	I	BENCH	CHD-002

6. Design constraints and budgets

- Passive measures must not depend on power (SRS 5.14 intent)
- Child-safe finish (CHD-002)

Budget	Allocation
Mass allocation	≤ 45 g (estimate 40 g)
Cost allocation	£3 (LED ring, tape); kit from owned fishing gear
Power	Beacon ≤ 0.3 W average

7. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	-	-	-	-	-	0
BENCH	1	-	4	-	-	5
POOL	1	-	-	-	-	1
LAKE	1	-	1	-	-	2

Ref	Open item
TBC-08	Beacon data level (IF-09)
REC-OI-1	Owner to supply the contact number for the labels

Subsystem Specification Simulation & test

Let the whole system be exercised, failed on purpose and rehearsed at home, first against a simulated boat and then with more and more real hardware in the loop, with the same software throughout.

Document	BOATY-SSS-SIM
Issue	Issue B (for review)
Date	28 September 2026
Status	For review by the project owner
Parents	SRS Issue D, ADD Issue C, ICD Issue B
Content	3 allocated SRS requirements → 24 subsystem requirements (18 M, 6 S, 0 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)
B	28 September 2026	Hardware-in-the-loop rigs L1-L3 added (SIM-D13 to D24). Test catalogue rebuilt with IDs and extended with FMEA-derived scenarios (BOATY-FMEA-001).	Claude, owner request

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Let the whole system be exercised, failed on purpose and rehearsed at home, first against a simulated boat and then with more and more real hardware in the loop, with the same software throughout.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• ArduPilot SITL (Rover, boat frame) • Camera stub (IF-03) • Launch scripts • Scenario runner and test suites • CI configuration • Rig L1: real computers + SITL • Rig L2: iron bird (real helm and power train) • Rig L3: water-tank thrust and weed rig	• The software under test (MCN, MCP services) • Pool and lake trials (per-subsystem and system test procedures)

1.3 Breakdown

Item	Description
SIM-1 Simulator	SITL + camera stub + launcher
SIM-2 Test suites	Unit, contract, scenario, FMEA-derived, LLM evaluation
SIM-3 CI	GitHub Actions workflows
SIM-4 Rig L1	Real Pi 5 panel, real Pi Zero + camera, real Wi-Fi; SITL on a laptop, reaching the Pi Zero UART through a USB-serial adapter
SIM-5 Rig L2 'iron bird'	Real FC, firmware and parameters, power board with key, ESCs and motors, GNSS, beacon, on a board; driven by the real Mission Control
SIM-6 Rig L3 tank	One pod in a water tub on a load cell, with current logging and real pond weed

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue C	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue B	Interface definitions (all listed in section 3 apply in full)

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-15 Site data	Mission Control	Conforms	\$5
IF-21 Simulation link	Mission Control	Owner	\$5

4. Parent requirements

Primary (3): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
SWE-004	M	The whole system shall run in simulation (helm simulator and simulated camera) on the Pi 5 or a laptop, with Mission...	SIM-D01, SIM-D02, SIM-D03, SIM-D04, SIM-D13
SWE-005	M	Each failsafe requirement (FS-001 to FS-013) shall have an automated simulation scenario.	SIM-D06, SIM-D07
SWE-007	S	Unit tests (pytest), linting (ruff) and type checking (mypy) shall run automatically on every push.	SIM-D11

5. Subsystem requirements

Requirements are numbered SIM-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Simulator

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D01	SITL shall run the same ArduPilot version as the helm, with a skid-steer boat frame and the controlled parameter file.	M	D	SIM	SWE-004, V-12
SIM-D02	The camera stub shall implement IF-03 exactly, serving test images geotagged from SITL.	M	T	SIM	SWE-004, IF-21
SIM-D03	One command shall start the simulator on a laptop or the Pi 5 at a given site's home.	M	D	SIM	SWE-004
SIM-D04	Mission Control shall run against the simulator with configuration changes only.	M	D	SIM	SWE-004, SWE-003
SIM-D05	A rehearsal mode shall run the physical panel against the simulator for crew practice.	S	D	SIM	OPS-012

5.2 Tests

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D06	Each FS-001 to FS-013 requirement shall have an automated scenario with explicit pass criteria (SC-01 to SC-13, section 6). FS-008 is proven on rig L2.	M	T	SIM	SWE-005
SIM-D07	Fault injection shall use SITL simulation parameters and network impairment, mapped per scenario.	M	T	SIM	SWE-005, IF-21
SIM-D08	Contract tests for IF-03 and IF-14 shall run against both the simulator and real hardware.	M	T	SIM	IF-14, IF-03
SIM-D09	The validator's adversarial suite (≥ 50 cases) and a property-based comparison against an independent geometry implementation shall run in CI (SC-31).	M	T	SIM	VAL-006
SIM-D10	The LLM evaluation set (≥ 30 instructions, including adversarial child phrasing) shall run on demand with a per-run cost cap (SC-33).	S	T	SIM	NLI-003, NLI-006

5.3 CI and reproducibility

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D11	Every push shall run ruff, mypy and unit tests. SITL scenarios shall run on pull requests and nightly.	S	I	SIM	SWE-007
SIM-D12	Tool and dependency versions (SITL commit, Python lock file) shall be pinned in the repository.	M	I	SIM	SWE-006

5.4 Hardware-in-the-loop rigs (Issue B)

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D13	Rig L1 shall connect SITL to the real Pi Zero 2W through a 3.3 V USB-serial adapter at 115200 baud on the Pi's UART, so the production router configuration is used unchanged. The real Pi 5 panel and Wi-Fi are used.	M	D	BENCH	SWE-004, IF-04, IF-01
SIM-D14	Rig L1 shall timestamp STOP from button edge to the SITL mode change with ≤ 10 ms resolution, over ≥ 100 trials (L1-03).	M	T	BENCH	FS-009, COM-003
SIM-D15	Rig L1 shall switch the Pi Zero's 5 V supply under script control for power-cut endurance testing (L1-08).	S	T	BENCH	IF-08

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D16	Rig L1 shall allow the sealed box with the Pi Zero to be heat-soaked at 30 °C (heat lamp or sun) while running a mission (L1-09).	S	T	BENCH	PWR-010
SIM-D17	Rig L2 shall mount the real FC (flashed firmware and controlled parameters), power board with fuse and magnetic key, ESCs, motors, GNSS with sky view, and beacon, driven by the real Mission Control over the real link.	M	D	BENCH	SAF-001, IF-05, IF-06, IF-07, IF-09
SIM-D18	Rig L2 shall support physical fault injection: GNSS unplug and data-line cut, AP off, bench-supply voltage ramp, ESC signal cut, key removal, FC power-cycle, and thrust steps.	M	T	BENCH	FS-001, FS-002, FS-004, FS-008, MOD-003, PWR-005
SIM-D19	Rig L2 shall record motor current, 5 V rails and ESC signals (logic analyser or scope), and detect motor stop within 50 ms (ESC telemetry or optical).	M	T	BENCH	FS-008, FS-009, PWR-005
SIM-D20	Rig L2 shall run with props removed in air, or with pods submerged in a covered tank. Removing the key is the emergency stop, and the adult-only rules apply.	M	I	BENCH	OPS-007, CHD-003
SIM-D21	Rig L3 shall measure one pod's thrust (± 0.05 N) and current, forward and astern, at 0-100% in 10% steps (L3-01).	M	T	BENCH	NAV-001, ENV-002, IF-06
SIM-D22	Rig L3 shall introduce real pond weed at the inlet and record whether the weed-shedding routine clears it (L3-02).	S	T	BENCH	ENV-005, FS-006
SIM-D23	Every rig test shall be recorded with date, software and firmware versions, parameter hash and results, and committed to the repository.	M	I	BENCH	SWE-006
SIM-D24	Every FMEA failure mode with severity ≥ 8 shall be exercised by at least one SIM, rig or bench test in the catalogue. The build checks this.	M	A	SIM	SAF-006, FS-013

6. Test catalogue

Every test has an ID, which the FMEA (BOATY-FMEA-001) and the test procedures reference. SC = simulator scenario; L1/L2/L3 = rig tests; B = bench; R = rehearsal; P = pool. Injection parameter names are fixed per SITL version (TBC-04) and kept in the scenario code.

6.1 Failsafe scenarios (one per FS requirement)

ID	Test	Injection / method	Pass criterion	Refs
SC-01	Battery drain to 35% then 15%	Simulated battery drain	RTL at 35%; alarm at 15%; reaches home	FS-001
SC-02	Link cut in STEERING	Drop UDP 14550	HOLD ≤ 2 s; RTL at 10 s (B4)	FS-002
SC-03	Link cut in AUTO	Drop UDP 14550	Mission continues; RTL at 60 s (B4)	FS-003, FM-31
SC-04	GNSS failure 5 s then restore	SITL GPS failure	Motors stop ≤ 3 s; RTL after 10 s healthy (B6)	FS-004, FM-01, FM-06
SC-05	Heavy drag at throttle $\geq 50\%$	SITL drag / wind	HOLD ≤ 5 s plus event	FS-005, FM-16
SC-06	Drag released after burst 2	SITL + B5	≤ 3 bursts; resumes mission	FS-006, FM-27
SC-07	Kill all MCP services mid-mission	Stop stub and B-services	Mission completes; RTL; HOLD at home	FS-007, FM-26
SC-08	Helm signal loss to ESCs	Covered by L2-10	See L2-10	FS-008
SC-09	STOP in every state	Panel emulation	Motors stop ≤ 1 s	FS-009
SC-10	Moisture flagged	Stub health = moisture	RTL ≤ 2 s	FS-010, FM-24
SC-11	GNSS loss during battery RTL	Combined	Motors-stopped (HOLD) wins	FS-011
SC-12	Log and announce audit	Log inspection over SC-01 to SC-11	Every event logged, announced ≤ 2 s	FS-012
SC-13	Single-fault sweep across mission phases	Scripted matrix of SC-20 to SC-39 faults	Never leaves the fence under power	FS-013

6.2 FMEA-derived simulator scenarios

ID	Test	Injection / method	Pass criterion	Refs
SC-20	GNSS glitch: 20-50 m jump for 2 s near the fence	SITL GPS glitch	No uncommanded exit; glitch rejected or HOLD	FM-02
SC-21	GNSS frozen: stale position while moving	SITL GPS freeze	Detected; motors stop ≤ 3 s	FM-03
SC-22	Compass offset 30° and 90°	SITL compass offset	Yaw fallback or HOLD; stays inside the fence	FM-04, FM-05
SC-23	Helm restart mid-mission	Restart SITL process	Motors stop; boots disarmed; alarm at Mission Control	FM-07
SC-24	Parameter differs from baseline	Change one failsafe parameter	Arming blocked; difference shown	FM-09
SC-25	Fence missing or disabled	Skip the fence upload	Arming refused	FM-10
SC-26	Site file lat/lon swapped; wrong site	Corrupt the site file	Linter and pre-arm both refuse	FM-11
SC-27	Breach on the far side of the island	Push the boat out with wind	RTL path avoids the exclusion	FM-13
SC-28	Persistent breach (wind pushing out)	Strong offshore wind	Motors stop by 30 s / 10 m	FM-14
SC-29	Single motor failure	Zero one output	Divergence watchdog → HOLD ≤ 20 s	FM-17
SC-30	Boat-service command fuzz	Random forbidden commands	All blocked by the filter	FM-27

ID	Test	Injection / method	Pass criterion	Refs
SC-31	Validator adversarial + property-based	Hypothesis vs independent geometry	No accepted mission crosses a boundary	FM-34
SC-32	Read-back corruption	Alter one item in transfer	GO stays disabled	FM-35, FM-47
SC-33	LLM evaluation with adversarial phrasing	Recorded instruction set	Schema-valid; must-decline declined	FM-36, FM-37
SC-34	Link degradation: 30/60/90% loss, 2 s latency	Network impairment	FS-002/003 behaviour holds; STOP latency logged	FM-39, FM-40
SC-35	Home sanity	Arm with GNSS unsettled / far from site home	Refused or warned before GO	FM-46
SC-36	Interrupted upload	Drop the link mid-transfer	No partial mission accepted	FM-47
SC-37	Foreign GCS heartbeat	Second system-255 source	C7 refuses to operate; alarm	FM-41
SC-38	First-motion heading check	Compass reversed	HOLD $\leq$ 10 s after start	FM-05, FM-18
SC-39	Capacity overstated by 30%	Wrong capacity parameter	Voltage backstop triggers RTL in time	FM-15

6.3 Rig L1: real computers, simulated boat

ID	Test	Injection / method	Pass criterion	Refs
L1-01	End-to-end mission, real computers	Voice $\rightarrow$ captain's log over real Wi-Fi and UART	Completes; all artefacts logged	SWE-004
L1-02	Panel self-test and state $\times$ button table	Real buttons	Matches ICD IF-12	FM-32, FM-33
L1-03	STOP latency, 100 trials	Button edge $\rightarrow$ SITL mode	$P99 \leq 0.5$ s	FM-40, FS-009
L1-04	Photo sync, 200 photos	Real Wi-Fi at 10 m	≤ 5 min; hashes verified	CAM-007
L1-05	UART load	Maximum telemetry + services	GCS heartbeat jitter < 0.5 s	FM-28
L1-06	Mission Control power pulled mid-mission	Unplug the Pi 5	B4 RTL at 60 s	FM-31
L1-07	Range walk on land	Walk the Pi Zero away	RSSI vs distance; flap behaviour as SC-34	FM-39
L1-08	Pi Zero power-cut $\times 50$	Switched 5 V	No filesystem damage	FM-26
L1-09	Sealed-box heat soak 30 °C	Heat lamp	No throttling for 60 min	PWR-010

6.4 Rig L2: iron bird

ID	Test	Injection / method	Pass criterion	Refs
L2-01	Motor direction and channel map	Props off	Left/right and sense correct	FM-18
L2-02	Arming gated on the key	Key in/out	Refused with key out; rail < 0.5 V	FM-22, MOD-003
L2-03	GNSS unplugged	Pull connector	EKF failsafe HOLD	FM-01
L2-04	GNSS data line cut, power on	Cut TX	Detected ≤ 3 s	FM-03
L2-05	Compass vs motor current	Motor-interference calibration	Interference $\leq 30\%$	FM-04
L2-06	Heading vs reference	Known bearing	Within 10°	FM-05
L2-07	FC power-cycle with motors running	Interrupt FC supply	Motors stop; boots disarmed	FM-07
L2-08	Thrust-step brownout	Full reverse $\rightarrow$ forward	5 V rails in regulation	FM-08, PWR-005
L2-09	Supply voltage ramp	Bench supply	Battery failsafe at thresholds	FM-15
L2-10	ESC signal cut and restore	Cut the signal lead	Stop ≤ 1 s; restart only from neutral	FM-19, FM-20, FS-008

ID	Test	Injection / method	Pass criterion	Refs
L2-11	STOP to motors stopped	Panel STOP	≤ 1 s	FS-009, FM-40
L2-12	Beacon patterns	Cycle states	Each state distinguishable	REC-004
L2-13	Key-switch short detection	Bridge the MOSFET	Checklist rail test flags it	FM-22

6.5 Rig L3: water tank

ID	Test	Injection / method	Pass criterion	Refs
L3-01	Thrust and current curves	Load cell, 0-100%	≥ 2 N forward per pod; current ≤ 8 A	NAV-001, IF-06
L3-02	Weed fouling trials	Real pond weed	Cleared in ≥ 2 of 3	FM-16, ENV-005
L3-03	Cruise power estimate	Thrust vs drag model	≤ 10 W at 1.0 m/s	PWR-009

6.6 Bench, rehearsal and pool items referenced by the FMEA

ID	Kind	Test	Pass criterion	Refs
B-01	BENCH	Prop guard probe	No blade contact	FM-21, FM-45
B-02	BENCH	BMS protection	BMS trips; fuse intact	FM-23
B-03	BENCH	Box dunk	No ingress	FM-24
B-04	BENCH	Harness pull and connector	No disconnect	FM-25
B-05	BENCH	Camera failure	Health reports fault; mission unaffected	FM-29
B-06	BENCH	Clock without GNSS	Timestamps flagged until GNSS time	FM-30
R-01	REHEARSAL	Operator contingency rehearsal	All section 3.5 scenarios rehearsed	FM-42, FM-43, FM-44, FM-12
P-01	POOL	Internet absent	Templates offered	FM-38

6.7 What the rigs verify earlier

Item	First verified (Issue A)	Now also by	Still needs
FS-009 STOP ≤ 1 s	BENCH / POOL	L1-03 (to helm), L2-11 (to motors stopped)	On-water confirmation (POOL)
COM-003 telemetry and latency	POOL	L1-03, L1-05	Range effects (LAKE)
CAM-007 / MCP-D16 photo sync	POOL	L1-04	-
MCN-D21 link quality display	POOL	L1-07	-
MCN-D22 manual drive	POOL	L2 (motors follow the joystick)	Handling (POOL)
NAV-001 bidirectional thrust	POOL	L2-01, L3-01	-
PRP-D02 thrust ≥ 4 N	BENCH (estimate)	L3-01 (measured)	-
PRP-D03/D04 cruise throttle and power	POOL	L3-03 (estimate from data)	Confirm (POOL)
PRP-D11 / ENV-005 weed	LAKE	L3-02 (real weed)	Confirm (LAKE)
FS-005/006 stuck and shedding	POOL	SC-05/06 + L3-02	Confirm (POOL)
FS-001 / FS-004 on real hardware	SIM only	L2-09, L2-03/04	-
REC-D03 beacon patterns	LAKE	L2-12 (patterns)	Visibility (LAKE)
PWR-010 thermal	BENCH	L1-09 under a real workload	-

The FMEA check (SIM-D24) confirms that every failure mode with severity ≥ 8 has at least one test here.

7. Design constraints and budgets

- Same parameter file as the real helm, plus SIM_* overrides
- Runs on a laptop or the Pi 5

- Rig L2 is run with props off, or in water with the tank covered. Removing the magnetic key is the emergency stop.

Budget	Allocation
Cost	£0 software (open source). Rig equipment, outside the Mk1 BOM: USB-serial adapter £3, load cell + HX711 £6, tub; bench power supply assumed owned.
LLM evaluation	API credit, capped per run
Mass / power	n/a

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	6	4	2	1	-	13
BENCH	7	2	2	-	-	11
POOL	-	-	-	-	-	0
LAKE	-	-	-	-	-	0

Ref	Open item
TBC-03 / V-12	SITL skid boat frame
TBC-04	Fault injection parameters per scenario
SIM-OI-1	Rig L2 needs the first parts order (FC, ESCs, motors, power parts). Rig L1 needs only a Pi Zero 2W and camera.